

1 Solution — GIAM 2.7

Problem 1

1.1 Problem

Determine the logical form of the following arguments. Use symbols to express that form and determine whether the form is **valid** or **invalid**. If invalid, determine the type of error. Comment on the **soundness** as well — in particular, whether any premises are questionable.

- (a) All who are guilty are in prison. George is not in prison. Therefore George is not guilty.
- (b) If one eats oranges one will have high levels of vitamin C. You do have high levels of vitamin C. Therefore you must eat oranges.
- (c) All fish live in water. The mackerel is a fish. Therefore the mackerel lives in water.
- (d) If you're lazy, don't take math courses. Everyone is lazy. Therefore no one should take math courses.
- (e) All fish live in water. The octopus lives in water. Therefore the octopus is a fish.
- (f) If a person goes into politics, they are a scoundrel. Harold has gone into politics. Therefore Harold is a scoundrel.

1.2 Summary of Answers

part	form	verdict	soundness
(a)	universal modus tollens	valid	unsound — premise false
(b)	affirming the consequent	invalid	—
(c)	universal modus ponens	valid	sound
(d)	universal modus ponens	valid	unsound — premise false
(e)	affirming the consequent	invalid	premises true, conclusion false

(f) universal modus ponens **valid** unsound — premise false

Table 1.1

Two questions per argument, asked separately: is the FORM valid, and are the PREMISES true?

(a) guilt and prison $\begin{array}{l} \forall x, G(x) \Rightarrow P(x) \\ \neg P(\text{George}) \\ \hline \therefore \neg G(\text{George}) \end{array}$ VALID — universal modus tollens <i>unsound: plenty of guilty people are not in prison</i>	(b) oranges and vitamin C $\begin{array}{l} O \Rightarrow V \\ V \\ \hline \therefore O \end{array}$ INVALID — affirming the consequent <i>peppers, broccoli and supplements also supply vitamin C</i>
(c) the mackerel $\begin{array}{l} \forall x, F(x) \Rightarrow W(x) \\ F(\text{mackerel}) \\ \hline \therefore W(\text{mackerel}) \end{array}$ VALID — universal modus ponens <i>and SOUND — the only good argument here</i>	(d) laziness and math $\begin{array}{l} \forall x, L(x) \Rightarrow \neg M(x) \\ \forall x, L(x) \\ \hline \therefore \forall x, \neg M(x) \end{array}$ VALID — universal modus ponens <i>unsound: "everyone is lazy" is plainly false</i>
(e) the octopus $\begin{array}{l} \forall x, F(x) \Rightarrow W(x) \\ W(\text{octopus}) \\ \hline \therefore F(\text{octopus}) \end{array}$ INVALID — affirming the consequent <i>both premises TRUE, conclusion FALSE</i>	(f) Harold the politician $\begin{array}{l} \forall x, P(x) \Rightarrow S(x) \\ P(\text{Harold}) \\ \hline \therefore S(\text{Harold}) \end{array}$ VALID — universal modus ponens <i>unsound: "all politicians are scoundrels" is a slur, not a fact</i>
The designed contrast: (c) and (e) share a premise but differ in direction Both start from $\forall x, F(x) \Rightarrow W(x)$ — "fish $\Rightarrow$ lives in water". Then: (c) affirms the antecedent: mackerel is a fish, so it lives in water. ✓ runs with the arrow (e) affirms the consequent: octopus lives in water, so it is a fish. ✗ runs against it <i>Living in water is necessary for being a fish, not sufficient — and the octopus is a mollusc.</i>	

Figure 1.1: Six panels, each showing an argument's symbolic form and two verdicts. Panel (a), guilt and prison: for-all x $G(x)$ implies $P(x)$, not P of George, therefore not G of George — VALID, universal modus tollens, but unsound since plenty of guilty people are not in prison. Panel (b), oranges and vitamin C: O implies V , V , therefore O — INVALID, affirming the consequent, since peppers, broccoli and supplements also supply vitamin C. The third panel, the mackerel, highlighted green: for-all x $F(x)$ implies $W(x)$, F of mackerel, therefore W of mackerel — VALID, universal modus ponens, and SOUND, the only good argument here. Panel (d), laziness and math: for-all x $L(x)$ implies not $M(x)$, for-all x $L(x)$, therefore for-all x not $M(x)$ — VALID, universal modus ponens, but unsound since 'everyone is lazy' is plainly false. Panel (e), the octopus, highlighted red: for-all x $F(x)$ implies $W(x)$, W of octopus, therefore F of octopus — INVALID, affirming the consequent, with both premises TRUE and the conclusion FALSE. Panel (f), Harold the politician: for-all x $P(x)$ implies $S(x)$, P of Harold, therefore S of

Harold — VALID, universal modus ponens, but unsound since ‘all politicians are scoundrels’ is a slur not a fact. A closing panel gives the designed contrast: the mackerel and octopus arguments share the premise ‘fish implies lives in water’, but the mackerel affirms the antecedent and runs with the arrow, while (e) affirms the consequent and runs against it. Living in water is necessary for being a fish, not sufficient — and the octopus is a mollusc.

1.3 The Six Arguments

1.3.1 (a) Guilt and prison — valid, unsound

Let $G(x)$ = " x is guilty," $P(x)$ = " x is in prison."

$$\frac{\forall x, G(x) \Rightarrow P(x) \quad \neg P(\text{George})}{\therefore \neg G(\text{George})}$$

Universal modus tollens — valid. But the first premise is false: many guilty people are never caught, and prisons hold some innocents besides. Valid form, worthless conclusion.

1.3.2 (b) Oranges and vitamin C — invalid

Let O = “you eat oranges,” V = “you have high vitamin C.”

$$\frac{O \Rightarrow V \quad V}{\therefore O}$$

Affirming the consequent — invalid (counterexample: O false, V true). Oranges are *sufficient* for high vitamin C, not *necessary*; peppers, broccoli and supplements do the job too.

1.3.3 (c) The mackerel — valid and sound

Let $F(x)$ = " x is a fish," $W(x)$ = " x lives in water."

$$\frac{\forall x, F(x) \Rightarrow W(x) \quad F(\text{mackerel})}{\therefore W(\text{mackerel})}$$

Universal modus ponens — valid, and both premises are true, so the argument is **sound**. This is the only genuinely good argument in the list.

1.3.4 (d) Laziness and math — valid, unsound

Let $L(x)$ = " x is lazy," $M(x)$ = " x should take math courses."

$$\frac{\forall x, L(x) \Rightarrow \neg M(x) \quad \forall x, L(x)}{\therefore \forall x, \neg M(x)}$$

Universal modus ponens, applied to an arbitrary x and then generalised — valid. Unsound: "everyone is lazy" is a sweeping falsehood, and the first premise is dubious advice rather than a fact.

1.3.5 (e) The octopus — invalid

$$\frac{\forall x, F(x) \Rightarrow W(x) \quad W(\text{octopus})}{\therefore F(\text{octopus})}$$

Affirming the consequent — invalid. This case is especially clean: **both premises are true** (all fish do live in water; the octopus does live in water) while **the conclusion is false** (an octopus is a mollusc). True premises with a false conclusion is a *proof* that the form cannot be valid.

1.3.6 (f) Harold the politician — valid, unsound

Let $P(x)$ = " x goes into politics," $S(x)$ = " x is a scoundrel."

$$\frac{\forall x, P(x) \Rightarrow S(x) \quad P(\text{Harold})}{\therefore S(\text{Harold})}$$

Universal modus ponens — valid. Unsound: “everyone who goes into politics is a scoundrel” is a cynical generalisation, not a truth.

1.4 Remark — (c) and (e) Are the Point of the Exercise

The two fish arguments share the *identical* first premise and differ only in which part of it the second premise attaches to:

$$\underbrace{F(\text{mackerel}) \Rightarrow W(\text{mackerel})}_{(c): \text{affirm the antecedent} \text{ — valid}}, \quad \underbrace{W(\text{octopus}) \not\Rightarrow F(\text{octopus})}_{(e): \text{affirm the consequent} \text{ — invalid}}.$$

“All fish live in water” says living in water is **necessary** for being a fish, not **sufficient**. Whales, octopuses, and jellyfish all satisfy the consequent without satisfying the antecedent. Running an implication backwards is exactly the converse error of Problem 2.2.9, and (e) is the cleanest possible demonstration of it because the premises really are true.

1.5 Remark — How to Prove a Form Invalid

Part (e) illustrates the standard technique. To show a form is **invalid**, exhibit an interpretation in which **every premise is true and the conclusion is false**. One such instance suffices, because validity means “no such interpretation exists.” Here the real world provides the counterexample for free — no artificial model needed.

Note the direction of the inference: a valid form can still have a false conclusion (if a premise is false), and an invalid form can still have a true conclusion (by luck). What is *impossible* is a valid form with true premises and a false conclusion. That is the entire content of validity, and it is what makes the octopus decisive.

1.6 Remark — Four Valid Arguments, Only One Worth Anything

Parts (a), (c), (d) and (f) are all valid, yet only (c) is any good. The other three run correct machinery on false inputs:

- (a) “all who are guilty are in prison” — plainly false.

- (d) “everyone is lazy” — plainly false.
- (f) “all politicians are scoundrels” — a prejudice stated as a universal.

Notice the shared shape of the failures: each unsound premise is a **sweeping universal** of the form $\forall x, \dots$, and each can be refuted by a single counterexample (Problem 2.3.5). Sweeping universals are where unsoundness usually enters an otherwise well-formed argument, which is why “is that premise *really* true of everyone?” is the most productive question to ask of an argument whose logic checks out.

1.7 Remark — Why Both Questions Must Be Asked

This problem set is built to force the two questions apart, and the six parts populate three of the four possible combinations:

	valid	invalid
all premises true	(c) — sound, conclusion guaranteed	(e) — conclusion happens to be false
some premise false	(a), (d), (f) — conclusion unsupported	(b) — doubly bad

Table 1.2

Only the top-left cell licenses believing the conclusion. Everything else leaves the conclusion unsupported, whether the fault lies in the form, the premises, or both. Checking validity is mechanical (Sections 2.6–2.7 give the catalogue of rules); checking premises requires knowing something about the world — which is why logic is a tool for reasoning, not a substitute for it.